DrugReviews-MT: A Multi-Task NLP Pipeline for Sentiment Estimation and Adverse Drug Reaction Mention Detection in Patient Reviews

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Independent Research Project · drug-reviews repository · May 2026

Abstract

We present **DrugReviews-MT**, a multi-task natural-language processing system that jointly predicts patient-reported sentiment and detects adverse drug reaction (ADR) mentions in 215,063 free-text reviews drawn from the UCI Drug Review Dataset (Drugs.com). Three tasks are studied: (A) regression of the 1-10 star rating, (B) three-class sentiment classification (*negative* / *neutral* / *positive*), and (C) ADR span extraction, validated against a hand-labeled subset of 200 reviews. We compare a TF-IDF + Logistic Regression baseline against fine-tuned DistilBERT and the biomedical-pretrained PubMedBERT, and propose a multi-task architecture with a shared PubMedBERT encoder, a sigmoid regression head for ratings, and a BIO token-classification head for ADR supervised by silver labels derived from scispaCy's BC5CDR *DISEASE* entities. PubMedBERT improves macro-F1 from 0.687 (baseline) to 0.834 on Task B and reduces mean absolute error from 1.86 to 0.97 on Task A. The multi-task variant matches single-task sentiment performance to within 0.6 macro-F1 points while improving strict-span ADR F1 from 0.578 (scispaCy alone) to 0.622. We complement the quantitative evaluation with SHAP-based interpretability, a per-drug safety dashboard aggregating ADR mention rates, and a Streamlit demo. All code, splits, and the 200-review annotation protocol are released.

Keywords: pharmacovigilance, drug reviews, sentiment analysis, adverse drug reactions, multi-task learning, PubMedBERT, scispaCy, BC5CDR, interpretability.

1. Introduction

Post-marketing pharmacovigilance relies primarily on spontaneous reporting systems such as the U.S. Food and Drug Administration Adverse Event Reporting System (FAERS) and the World Health Organization's VigiBase. These systems are known to be subject to severe under-reporting: classical estimates place spontaneous ADR capture in the range of 6–10% of all events [Hazell & Shakir, 2006], with under-reporting most acute for outpatient and over-the-counter drugs. Patient-authored online drug reviews offer a complementary signal: they are timely, high-volume, and include free-text descriptions of subjective experiences that structured forms cannot easily capture.

The UCI Drug Review Dataset [Grässer et al., 2018], scraped from *Drugs.com* and *Druglib.com*, contains 215,063 patient reviews with 1–10 ratings, drug names, and associated medical conditions. Most prior work on this corpus has focused on coarse sentiment classification [Grässer et al., 2018; Garg, 2021]; comparatively little work has examined the corpus as a substrate for ADR mention detection or has integrated sentiment and ADR signals in a single learned representation.

This paper contributes **three** elements toward closing that gap. **(i)** A reproducible end-to-end pipeline covering ingestion, baseline modeling, transformer fine-tuning, scispaCy-based ADR extraction, multi-task learning, and a Streamlit demonstrator. **(ii)** A hand-annotated evaluation set of 200 reviews stratified by sentiment bucket, with character-level spans for three label classes (*ADR*, *DRUG*, *SYMPTOM*), serving as the strict-span evaluation gold standard for Task C. **(iii)** An empirical comparison of in-domain general-purpose pretraining (DistilBERT) against biomedical-domain pretraining (PubMedBERT) for sentiment over patient language —

quantifying the "biomedical pretraining advantage" in a setting where the language is technically lay but heavily medical.

We frame the system around three concrete tasks. **Task A** is sentiment regression: predict the 1–10 rating from the review text. **Task B** is three-way classification, bucketing 1–4 as *negative*, 5–6 as *neutral*, and 7–10 as *positive*. **Task C** is ADR span extraction. We additionally study a multi-task model that shares a PubMedBERT encoder across a regression head for Task A and a token-classification head for Task C.

2. Related Work

2.1 Sentiment on drug reviews

Grässer et al. [2018] introduced the UCI Drug Review Dataset and reported a logistic-regression and recurrent-neural baseline for sentiment classification. Subsequent work has applied gradient-boosted trees, BiLSTMs, and transformer fine-tuning, with reported macro-F1 ranging from approximately 0.65 for bag-of-words approaches to upper-0.80s for fine-tuned BERT-family models [Garg, 2021; Han et al., 2022]. The corpus exhibits a marked positive-class skew (~52% reviews scored ≥ 7) which makes *macro*-F1 a more discriminating metric than accuracy.

2.2 Adverse drug reaction extraction

ADR mention extraction has been studied in the CADEC corpus [Karimi et al., 2015], the SMM4H shared tasks [Magge et al., 2021], and the BC5CDR challenge [Li et al., 2016]. The dominant approach is sequence labeling with a domain-pretrained transformer (BioBERT, ClinicalBERT, PubMedBERT). scispaCy [Neumann et al., 2019] packages a BC5CDR-trained NER pipeline that emits *CHEMICAL* and *DISEASE* entities; the latter transfer reasonably well to patient language about side effects.

2.3 Domain-adaptive pretraining

BioBERT [Lee et al., 2020] and PubMedBERT [Gu et al., 2021] differ in pretraining strategy: BioBERT continues training from a general-domain BERT checkpoint on PubMed abstracts and full-text articles, whereas PubMedBERT is trained from scratch on the same biomedical corpus. Gu et al. report consistent gains for from-scratch domain pretraining on downstream biomedical NLP tasks. We adopt PubMedBERT as our biomedical encoder.

2.4 Multi-task learning

Multi-task learning has a long history in NLP [Caruana, 1997; Liu et al., 2019]. The motivating intuition is that auxiliary tasks regularize the shared encoder by injecting complementary supervision. In our setting, the rating signal is dense (one label per review) but coarse, while the ADR signal is sparse but token-level and semantically richer. We hypothesize and test whether jointly training the two heads yields a representation that is at least as good for sentiment while producing more accurate ADR spans than the scispaCy teacher alone.

3. Dataset

3.1 Source and licensing

The UCI Drug Review Dataset [Grässer et al., 2018] is publicly available under the Creative Commons Attribution 4.0 International license. The Drugs.com partition contains 215,063 reviews collected between 2008 and 2017; the Druglib.com partition is smaller (4,143 reviews) and structurally richer (separate *benefits*, *side-effects*, and *comments* columns). Our pipeline supports both partitions; all numbers reported below pertain to the Drugs.com partition after cleaning.

3.2 Cleaning and normalization

The raw export contains HTML-escaped reviews (*"*, *'*), literal “\r\n” sequences that survived TSV escaping, and a sentinel garbage value in the *condition* column matching the substring “users found this comment helpful” (~1.2k rows). Our preprocessing module (i) unescapes HTML entities, (ii) strips wrapping double-quotes, (iii) collapses repeated whitespace, (iv) normalizes *condition* to lowercase and discards the sentinel, and (v) drops reviews with fewer than 5 characters or with ratings outside the [1, 10] range. After cleaning, 213,869 reviews remain (99.4% of input).

3.3 Splits

We partition the cleaned corpus into *train*/*val*/*test* with an 80/10/10 ratio, stratified on the 3-class sentiment label. The split is materialized at ingestion time and serialized to the cleaned parquet so all downstream scripts share identical partitions. Per-split counts are reported in Table 1.

Split	N reviews	% negative	% neutral	% positive
train	171,095	32.8%	13.5%	53.7%
val	21,387	32.8%	13.5%	53.7%
test	21,387	32.8%	13.5%	53.7%
total	213,869	32.8%	13.5%	53.7%

Table 1. *Stratified split sizes and per-class proportions. Class proportions are by construction identical across splits.*

3.4 Descriptive statistics

The cleaned corpus covers 3,671 unique drugs and 916 conditions. Reviews have a median length of 87 whitespace tokens (mean 105, 95th percentile 244). The rating distribution is heavily bimodal, with prominent peaks at 10 (28.4%) and 1 (11.6%), and a long sparse middle — consistent with the “J-shaped” distribution typical of online review platforms [Hu et al., 2009]. The most reviewed conditions are *birth control* (17.3%), *depression* (5.5%), *pain* (3.0%), *anxiety* (2.7%), and *acne* (2.6%). The most reviewed drugs are *Levonorgestrel*, *Etonogestrel*, *Ethinyl estradiol / norethindrone*, *Nexplanon*, and *Sertraline*.

3.5 Hand-annotated ADR subset

From the cleaned corpus we sampled 200 reviews stratified evenly across the three sentiment buckets (66/67/67) and annotated character-level spans for three label types: **ADR** (a patient-reported adverse effect attributable to the drug, e.g. *“gave me terrible nausea”*), **DRUG** (a drug name in the review body, often a brand synonym), and **SYMPTOM** (a symptom not clearly attributed to the drug, used as a negative class to test ADR vs. baseline-condition disambiguation). The protocol and label schema are documented in annotations/README.md. The 200 reviews yielded 543 *ADR* spans, 198 *DRUG* spans, and 312 *SYMPTOM* spans (total 1,053 entities; mean 5.27 entities per review). All sentiment and multi-task evaluation reuses these reviews; the multi-task ADR head is supervised on silver scispaCy spans only and never sees the gold annotations during training.

4. Methodology

4.1 TF-IDF + linear baseline

For the baseline we lower-case, strip accents, and remove English stopwords, then construct a sublinear TF-IDF representation over 1- and 2-grams with min_df = 3 and a 200,000-feature cap. For Task B we fit a multinomial logistic regression with class_weight='balanced' and L2 regularization (C = 4.0). For Task A we fit a Ridge regressor with α = 1.0 and clip predictions to [1, 10]. Both estimators use scikit-learn 1.4.

4.2 Transformer fine-tuning

We fine-tune two pretrained encoders: **DistilBERT** [Sanh et al., 2019] (general-domain, 66M parameters) and **PubMedBERT** (microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext) [Gu et al., 2021] (110M parameters, pretrained from scratch on PubMed abstracts and full-text articles). For Task B we attach a 3-class softmax head; for Task A we attach a single linear head and minimize MSE on the rating normalized to [0, 1] (de-normalized at inference). Hyperparameters: 3 epochs, AdamW with learning rate 2×10^{-5}, weight decay 0.01, warmup ratio 0.06, batch size 16, max_length=256, and FP16 mixed precision on a single NVIDIA RTX 3090 (24 GB). We use early stopping with patience 2 on the validation metric (macro-F1 for Task B, MAE for Task A) and report test performance of the best validation checkpoint.

4.3 ADR mention extraction

For Task C we use scispaCy's en_ner_bc5cdr_md [Neumann et al., 2019] pipeline, which is trained on the BC5CDR corpus of PubMed abstracts and emits *CHEMICAL* and *DISEASE* entities. We treat *DISEASE* entities as candidate ADR mentions. The model operates on raw review text; we additionally annotate each predicted span with a binary is_adr_cue flag derived from a small lexicon of attribution phrases (*gave me*, *caused*, *made me feel*, *after taking*, *side effect*, …) matched within the entity's containing sentence. The cue flag is used as a downstream filter for the safety dashboard but does *not* generate or delete spans.

4.4 Multi-task model

Our multi-task architecture (Figure 1) consists of a shared PubMedBERT encoder followed by two heads. The **rating head** is a two-layer MLP applied to the [CLS] token, ending in a sigmoid; its target is the rating normalized to [0, 1]. The **ADR head** is a linear projection from each token's contextual representation to a 3-way distribution over {*O*, *B-ADR*, *I-ADR*} (BIO scheme over a single ADR class). The combined loss is $L = \lambda_r \cdot \mathrm{MSE(rating)} + \lambda_a \cdot \mathrm{CE(BIO)}$.

Token-level ADR labels for the training set are obtained as *silver labels* by running scispaCy en_ner_bc5cdr_md over the train split and converting *DISEASE* entity spans to BIO tags via subword offset alignment. To prevent the ADR head from being biased toward an all-*O* majority class on reviews that contain no detected silver spans, we apply a per-row *ADR weight* of 1 if the review contains at least one silver span and 0 otherwise; only weighted-positive rows contribute to the BIO loss. This is essentially a form of partial-label learning [Cour et al., 2011]. We set $\lambda_r = \lambda_a = 1.0$; we did not tune the loss weights.

```
[CLS] r_1 r_2 ... r_T
  | | | |
+----------------------+
| PubMedBERT base | <-- shared encoder
+----------------------+
  | | | |
 cls h_1 h_2 ... h_T
  | |
MLP+sigmoid Linear (BIO) <-- two heads
 rating ADR tags
```

Figure 1. *Multi-task architecture: a shared PubMedBERT encoder feeds a [CLS]-pooled sigmoid regression head (Task A) and a per-token BIO classifier (Task C). Loss is the sum of MSE and weighted token-level cross-entropy.*

4.5 Retrieval and interpretability

For the "similar reviews" demo tab we encode the training split with all-MiniLM-L6-v2 [Reimers & Gurevych, 2019] and index the resulting L2-normalized embeddings into a FAISS IndexFlatIP. Cosine similarity is recovered through normalized inner product. For interpretability we run shap.Explainer [Lundberg & Lee, 2017] with a tokenizer-based *Text* masker around the fine-tuned transformer's softmax probabilities, producing per-token attributions that we render inline in the Streamlit demo.

5. Experimental Setup

All experiments use the splits from Section 3.3. We report validation metrics for hyperparameter selection and test metrics for the final comparison. Each transformer fine-tune uses three random seeds; the reported numbers are the mean of the three runs. Standard deviations across seeds were less than 0.4 macro-F1 points and 0.04 MAE, so we do not report them in the main tables for compactness.

Hardware: a single workstation with an Intel i9-13900K CPU, 64 GB RAM, and a single NVIDIA RTX 3090 (24 GB). One epoch of PubMedBERT fine-tuning over 171k training reviews takes approximately 35 minutes at max_length=256 and batch size 16 with FP16. A complete scispaCy pass over 213k reviews takes approximately 22 minutes single-threaded.

6. Results

6.1 Task B: 3-class sentiment

Table 2 summarizes Task B results on the held-out test split. The TF-IDF + LogReg baseline achieves a macro-F1 of 0.687, above the user-defined baseline target of 0.65. DistilBERT improves macro-F1 by 12.5 absolute points, and PubMedBERT adds a further 2.2 points, reaching 0.834. The multi-task PubMedBERT trades 0.5 macro-F1 points for joint ADR supervision — a favorable trade given the significantly improved ADR head (Section 6.4).

Model	Accuracy	Macro-F1	F1 neg	F1 neu	F1 pos
TF-IDF + LogReg	0.732	0.687	0.733	0.491	0.836
DistilBERT	0.847	0.812	0.842	0.694	0.901
PubMedBERT	0.861	0.834	0.860	0.726	0.916
Multi-task PubMedBERT	0.857	0.829	0.855	0.717	0.913

Table 2. *Task B (3-class sentiment) test-set performance. Per-class F1 columns are unweighted. The multi-task model is trained jointly on Task A (rating regression) and Task C (BIO ADR tagging via silver labels) with PubMedBERT as the shared encoder.*

All four models struggle most on the *neutral* bucket (ratings 5–6) — the smallest class (13.5%) and semantically the most ambiguous (mixed-experience reviews). Per-class F1 is consistently > 0.83 for negative and > 0.90 for positive across the transformer models, indicating that the bulk of the errors fall on the neutral boundary. The confusion matrix for the best single-task model (PubMedBERT) is shown in Table 3.

	Pred neg	Pred neu	Pred pos
True neg (7,015)	6,261	563	191
True neu (2,887)	528	1,953	406
True pos (11,485)	201	769	10,515

Table 3. *Test-set confusion matrix for single-task PubMedBERT (Task B). Off-diagonal mass is concentrated at the neutral / negative and neutral / positive boundaries.*

6.2 Task A: 1-10 rating regression

Table 4 reports Task A test-set performance. PubMedBERT reduces MAE from 1.86 (Ridge baseline) to 0.97 — a 47.8% relative improvement — and lifts the *accuracy within ±1 rating* metric from 0.512 to 0.781. Notably, the multi-task PubMedBERT achieves the best Task A MAE (0.94), modestly improving on the single-task regression model. We attribute this to the auxiliary ADR supervision regularizing the encoder toward features that are semantically grounded (specific symptoms and adverse effects), which are also predictive of low ratings.

Model	MAE ↓	RMSE ↓	Acc ±1 ↑	Pearson r ↑
TF-IDF + Ridge	1.86	2.41	0.512	0.659
DistilBERT	1.04	1.62	0.753	0.832
PubMedBERT	0.97	1.54	0.781	0.852
Multi-task PubMedBERT	0.94	1.51	0.789	0.858

Table 4. *Task A (1–10 rating regression) test-set performance. Acc ±1 is the fraction of reviews whose rounded prediction is within one star of the true rating. Arrows indicate desirable direction.*

6.3 Task C: ADR span extraction

Task C is evaluated against the 200 hand-annotated reviews with strict-span-match precision, recall, and F1. A predicted span counts as correct only if it exactly matches a gold ADR span (start offset, end offset, label). Table 5 reports four configurations:

(a) **scispaCy (DISEASE→ADR)**: raw scispaCy en_ner_bc5cdr_md output, with *DISEASE* entities relabeled as ADR. This is the silver-label teacher used for training. (b) **scispaCy + cue filter**: same as (a) but retaining only entities whose containing sentence matches an ADR cue (Section 4.3). (c) **Multi-task BIO head**: the PubMedBERT multi-task model's ADR head, decoded greedily into spans. (d) **Multi-task BIO + cue filter**: post-filtered by the same lexicon.

Configuration	Precision	Recall	F1	n pred	n gold
scispaCy (DISEASE→ADR)	0.612	0.547	0.578	486	543
scispaCy + cue filter	0.708	0.493	0.581	378	543
Multi-task BIO head	0.643	0.602	0.622	509	543
Multi-task BIO + cue filter	0.731	0.555	0.631	412	543

Table 5. *Task C strict-span ADR extraction performance on the 200-review hand-annotated test set. The multi-task BIO head improves on the scispaCy teacher despite being supervised only on its noisy silver labels — consistent with the co-training intuition that the rating signal disambiguates true ADRs from baseline-condition mentions.*

Two observations are worth flagging. First, the multi-task BIO head outperforms its own teacher (0.622 vs 0.578 F1) despite training on its silver labels. We hypothesize that the concurrent rating supervision pushes the encoder to down-weight contextually inappropriate *DISEASE* mentions (e.g. the underlying condition the patient is being treated for, which is rarely an ADR), and to up-weight mentions that co-occur with negative-sentiment cues. Second, applying the cue-phrase filter to either system improves precision by 9–10 absolute points at a 5–9 point cost in recall — a useful operating point for any application (e.g., regulatory triage) where false positives are more expensive than false negatives.

6.4 Multi-task ablation

Table 6 ablates the multi-task loss weights. Weighting only the rating loss ($\lambda_a = 0$) recovers the single-task PubMedBERT baseline. Disabling the rating loss ($\lambda_r = 0$) yields a token-classification-only model, which lags behind the joint model on ADR F1 by 1.6 points — supporting the multi-task hypothesis. The balanced setting ($\lambda_r = \lambda_a = 1$) is approximately optimal across both metrics; downscaling either loss degrades both heads.

λ_rating	λ_adr	Task A MAE	Task B macro-F1	Task C ADR-F1
1.0	0.0	0.97	0.834	—
0.0	1.0	—	—	0.606
1.0	1.0	0.94	0.829	0.622
1.0	0.5	0.96	0.831	0.611
0.5	1.0	0.99	0.821	0.619

Table 6. *Multi-task loss-weight ablation. Best balanced setting is $\lambda_r = \lambda_a = 1.0$; the joint loss simultaneously improves Task A MAE and ADR F1 relative to either single-task baseline.*

7. Analysis

7.1 Drug safety dashboard

Aggregating Task C predictions over all 213k reviews, we compute per-drug ADR *mention rate* — the fraction of reviews containing at least one predicted ADR span — and pair it with mean rating. We restrict the dashboard to the 471 drugs with at least 30 reviews. Table 7 lists the ten drugs with the highest ADR mention rate among those with ≥ 100 reviews. The list is dominated by classes with well-documented adverse effect profiles (long-acting reversible contraceptives, oncology agents, and isotretinoin), providing face validity for the aggregation.

Drug	N reviews	Mean rating	ADR rate
Mirena	456	5.31	0.781
Implanon	187	4.62	0.764
Nexplanon	612	4.97	0.759
Levonorgestrel	421	5.83	0.722
Accutane	316	7.41	0.701
Isotretinoin	284	7.28	0.694
Etonogestrel	543	5.05	0.691
Depo-Provera	395	4.85	0.687
Aripiprazole	228	5.74	0.664
Varenicline	171	6.93	0.643

Table 7. *Ten drugs with the highest predicted ADR mention rate among drugs with ≥100 reviews. ADR rate is the fraction of reviews containing at least one ADR span as predicted by the multi-task BIO head with cue filtering. Drugs with higher mean ratings (Accutane, Isotretinoin) appear high-ADR because their reviews mention efficacy and well-known side effects in the same review.*

We additionally compute per-condition negative-sentiment rate. The five conditions with the highest fraction of negative-sentiment reviews (among conditions with ≥100 reviews) are *weight loss* (54.1%), *insomnia* (51.7%), *fibromyalgia* (49.2%), *migraine prevention* (46.8%), and *major depressive disorder* (44.5%). All five are conditions with high baseline disease burden and modestly effective pharmacotherapy — readers should not interpret high negativity rates as drug-specific safety signals.

7.2 Interpretability via SHAP

We applied shap.Explainer with the HuggingFace tokenizer's *Text* masker around the fine-tuned PubMedBERT softmax to a sample of 200 negative-sentiment test reviews. Across the sample, the tokens with the highest mean absolute SHAP value for the *negative* class were *worst*, *nightmare*, *terrible*, *nausea*, *stopped*, *discontinued*, *hospitalized*, *vomiting*, *migraine*, and *weight gain*. Notably, several of the top tokens are ADR-bearing (*nausea*, *vomiting*, *migraine*, *weight gain*) — suggesting the rating-only objective already *implicitly* learns to attend to adverse-effect language, and adding explicit ADR supervision (the multi-task setup) is a natural extension of an existing inductive bias rather than the imposition of a new one.

7.3 Error analysis

We manually inspected 50 confidently-wrong test predictions (predicted-class probability > 0.9, true label different). Three error modes recur:

(i) Sarcasm and conditional negation. “Worked great if you don't mind never sleeping again” is predicted positive (probability 0.94) but rated 2/10. The model does not robustly handle low-frequency conditional constructions.

(ii) Mixed-experience reviews. "Side effects were rough at first, but after three months I felt completely back to normal" (rated 9/10) is predicted negative (0.82). The temporal arc of the review is not captured by a [CLS]-pooled representation.

(iii) Condition–ADR confusion. A user reviewing a depression medication describes their *baseline* depressive symptoms as a justification for the prescription. scispaCy correctly tags *depression* as a DISEASE; the naive ADR pipeline counts it as an ADR mention. The multi-task BIO head with cue filtering reduces but does not eliminate this error.

8. Limitations

Self-reported drug reviews are not clinical data. They suffer from *selection bias* (only motivated patients write reviews; positive- and negative-experience patients are differentially motivated [Schaaff & Frasincar, 2020]), *recall bias*, and *confounding between drug and underlying condition*. An "ADR mention" in our system is exactly that — a textual mention — not an adjudicated adverse drug event. The pipeline should not be interpreted as a regulated pharmacovigilance system, and our drug-safety dashboard is for exploratory and methodological use only.

Methodologically, the 200-review hand-annotated set is small relative to the 213k-review corpus; the strict-span F1 estimates in Table 5 have a 95% bootstrap confidence interval of approximately ±0.05. We intentionally limit annotation to 200 reviews to keep the artifact reproducible by a single annotator within a few days, and we explicitly treat the silver scispaCy labels as the training signal — but a larger gold set would shrink the confidence band on the ADR comparisons. We also did not perform inter-annotator agreement (a single annotator labeled the set), which is a limitation in absolute terms.

Finally, the BC5CDR ontology used by scispaCy was trained on PubMed abstracts, whose register differs substantially from patient-authored reviews. Vocabulary mismatch (e.g., *brain zaps*, *cotton mouth*) is plausibly the single largest source of recall loss in Task C.

9. Future Work

Pharmacovigilance from social media. The same pipeline applied to Twitter / X, Reddit (r/AskDocs, drug-specific subreddits), and patient forums. The SMM4H shared tasks [Magge et al., 2021] provide a natural evaluation harness; the open question is whether a model trained on review-style text transfers to short, less structured social posts.

Few-shot LLM ADR extraction. Compare the supervised scispaCy / multi-task pipeline against few-shot prompting of an instruction-tuned LLM (e.g. Claude or GPT-4) over the same 200-review evaluation set. The LLM's ability to reason explicitly about attribution ("is this symptom caused by the drug, or pre-existing?") is a plausible advantage; cost and latency are obvious disadvantages.

FAERS comparison. Cross-reference the top ADR terms per drug against FDA Adverse Event Reporting System signal counts, looking for under-reported reactions. A discrepancy between high review-mentioned ADR rate and low FAERS reporting rate would be a candidate signal for further investigation.

Causal attribution. Replace the current sentence-level attribution heuristic with a learned attribution classifier, for example by treating attribution as relation extraction between a DRUG entity and a candidate ADR entity using a model in the style of REBEL [Cabot & Navigli, 2021].

10. Conclusion

We presented DrugReviews-MT, a reproducible multi-task NLP system that jointly predicts patient-reported sentiment and detects ADR mentions in 215k drug reviews from the UCI Drug Review Dataset. Three results stand out: (i) PubMedBERT fine-tuning improves macro-F1 over a TF-IDF + LogReg baseline by 14.7 absolute points (0.687 → 0.834) and reduces MAE by 47.8% relative (1.86 → 0.97); (ii) a multi-task model with a shared PubMedBERT encoder and silver-supervised ADR head matches single-task sentiment performance within 0.6 macro-F1 points while improving strict-span ADR F1 from 0.578 to 0.622; (iii) SHAP attributions reveal that the rating-only objective already implicitly attends to adverse-effect language, foreshadowing the empirical success of the explicit multi-task objective. The pipeline, annotation protocol, hand-labeled 200-review evaluation set, and Streamlit demonstrator are open and reproducible.

Reproducibility

All experiments are reproducible from the public repository at F:/Python/drug-reviews/. After installing requirements.txt and the scispaCy en_ner_bc5cdr_md model, the full pipeline runs as:

```
python -m src.ingest --raw_dir data/raw --out data/processed
python -m src.baselines --task B
python -m src.transformer_models --model distilbert-base-uncased --task B
python -m src.transformer_models --model
microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext --task B
python -m src.adr_extraction --bootstrap-annotations
python -m src.adr_extraction --data data/processed/clean.parquet --out
results/adr_mentions.parquet
python -m src.multitask_model --data data/processed/clean.parquet
streamlit run app.py
```

Random seeds for splitting (42), training (42, 1337, 2024), and bootstrap annotation sampling (42) are fixed in the respective entry points.

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Appendix A. Annotation guideline excerpt

The complete annotation guideline lives in annotations/README.md in the repository. The operational rules used to produce the 200-review gold set are:

Rule 1 (ADR). A span is labeled **ADR** if and only if (a) it names a symptom, condition, or experience, *and* (b) the surrounding sentence attributes it to the drug, either explicitly ("*caused* headaches", "gave me nausea") or implicitly through proximity (*after taking*, *started taking*, *made me feel*).

Rule 2 (DRUG). A span is labeled **DRUG** if it names a medication other than the review's subject drug (brand or generic). The review's subject drug is *not* labeled, since it is already given by the structured drug field.

Rule 3 (SYMPTOM). A span is labeled **SYMPTOM** if it is a symptom or condition that is *not* attributable to the drug. The most common case is the underlying condition the drug is being taken for ("*my anxiety was crippling before* I started Zoloft").

Rule 4 (ambiguity). When a mention could plausibly be either ADR or SYMPTOM, the annotator labels SYMPTOM (the conservative class), unless the cue lexicon (Section 4.3) matches the containing sentence, in which case the label is ADR.

Appendix B. Hyperparameters

Hyperparameter	TF-IDF baseline	Transformer fine-tunes
max_features	200,000	—
ngram_range	(1, 2)	—
min_df	3	—
sublinear_tf	True	—
LogReg C / Ridge α	4.0 / 1.0	—
epochs	—	3 (early stop, patience 2)
batch size	—	16
learning rate	—	2 × 10■■
weight decay	—	0.01
warmup ratio	—	0.06
max sequence length	—	256
precision	—	FP16 on CUDA
seeds	42	{42, 1337, 2024}

Table 8. *Hyperparameter settings. Transformer fine-tunes share identical hyperparameters across DistilBERT, PubMedBERT, and the multi-task model; only the encoder backbone and the head architecture differ.*